International Journal of Forecasting 42 (2026) 1215–1225

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Stealing accuracy: Predicting day-ahead electricity prices with temporal hierarchy forecasting (THieF)☆

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ARTICLE INFO

Article history:
Dataset link: https://github.com/lipiecki/thief

Keywords:
Electricity price
Temporal hierarchy forecasting (THieF)
Forecast reconciliation
Regression
Neural network
Gradient-boosted decision tree
Transformer

ABSTRACT

We introduce the concept of *temporal hierarchy forecasting* (THieF) in predicting day-ahead electricity prices and show that reconciling forecasts for hourly products and 2- to 24-hour blocks can significantly (up to 13%) improve accuracy at all levels. These results remain consistent throughout a challenging 4-year test period (2021–2024) in the German and Spanish power markets and across model architectures, including linear regression, shallow feedforward neural networks, gradient-boosted decision trees, and a state-of-the-art, pretrained transformer. Given that (*i*) trading of block products is becoming more common and (*ii*) the computational cost of reconciliation is comparable to that of predicting hourly prices alone, we recommend using it in daily forecasting practice.

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1. Introduction

Operational decisions often require tailored short-term forecasts with varying levels of detail and granularity (Petropoulos et al., 2022). For instance, models for hourly products in wholesale electricity markets can use different information sets than those for average daily prices (Maciejowska & Weron, 2016; Weron, 2014). These forecasts may not align, leading to suboptimal decisions. To cope with this, the forecasts at each temporal level of the hierarchy should be reconciled to ensure coherence.

The last decade has seen an unprecedented growth in interest in forecast reconciliation (see Athanasopoulos et al., 2024, for a recent review) and the introduction of *temporal hierarchy forecasting* (THieF; Athanasopoulos et al., 2017). The latter can be applied to any time series via non-overlapping temporal aggregation – the predictions computed at all levels of the hierarchy are combined to yield temporally reconciled, accurate, and robust forecasts. The concept is new in energy forecasting, and there are only a handful of publications on predicting electric load (Møller et al., 2024; Nystrup et al., 2020) or wind (English & Abolghasemi, 2024; Jeon et al., 2019; Sharma et al., 2024) and solar (Di Fonzo & Girolimetto, 2023; Yang et al., 2017) generation. More importantly, temporal hierarchies have not yet been applied to *electricity price forecasting* (EPF). The only related study predicts aggregated demand and supply curves by exploiting their intrinsic hierarchical structure (Ghelasi & Ziel, 2024), but is not concerned with temporal-hierarchy forecasting or EPF.

To fill this gap and provide market participants with a universal tool to improve the predictions of hourly, block, and baseload prices (i.e., daily average prices), we conduct an extensive study involving four classes of models and two 4-year test periods (2021–2024) from two major European power markets: Germany and Spain. We use a variety of base forecasting methods with increasing complexity and diverse modeling assumptions, ranging from a parsimonious expert model built on classical linear regression (Serafin & Weron, 2025; Ziel & Weron, 2018) to Amazon's state-of-the-art tabular foundation model

☆ The numerical results presented in this manuscript were reproduced by the Reproducibility team on 27 March 2026.

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https://doi.org/10.1016/j.ijforecast.2026.03.003
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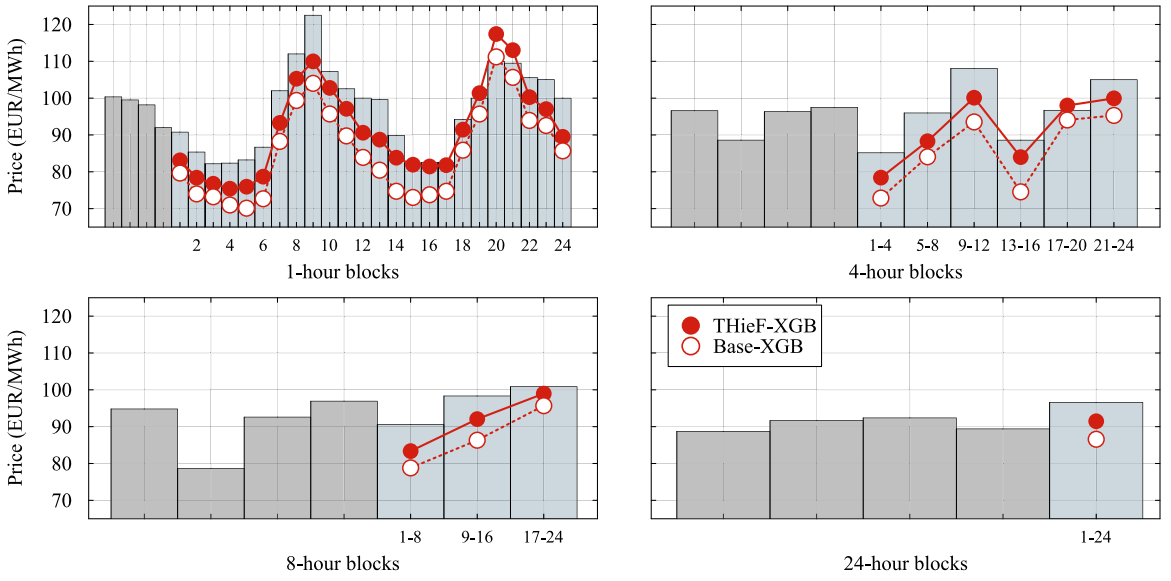


Fig. 1. Stylized example of the impact of THieF on forecasts at four hierarchy levels (1H, 4H, 8H, 24H). Filled red circles denote reconciled (→ THieF-XGB) and hollow circles unreconciled forecasts (→ Base-XGB) obtained using eXtreme Gradient Boosting (see Section 4.4). Bars represent actual block prices, with light blue indicating the target day, i.e., Sunday, 7.02.2021, in the German EPEX-DE market.

based on a 12-layer 72 million-parameter transformer architecture (Zhang et al., 2025).

To the best of our knowledge, we are the first to use pretrained foundation models not only for EPF but also for temporal hierarchies. As such, we complement the literature that has focused on evaluating the benefits of THieF with conventional, relatively small, base forecasting methods. We find that temporal reconciliation leads to significant improvements in mean absolute error (MAE) and root mean squared error (RMSE) across all model classes and hierarchy levels considered: up to 5.5% for hourly prices and 13.4% for baseload prices.

The remainder of the paper is structured as follows. In Section 2, we briefly review the THieF approach and discuss its application to electricity price forecasting. Next, in Section 3 we introduce the datasets and in Section 4 we explain how the base forecasts for the 24 hourly prices are determined using four classes of models: linear regression (→ Section 4.2), committee machines of shallow feedforward neural networks (→ Section 4.3), gradient-boosted decision trees (→ Section 4.4), and a state-of-the-art, pretrained transformer (→ Section 4.5). In Section 5 we discuss the results obtained. Finally, in Section 6, we conclude and speculate on how our approach can be extended in future studies.

2. Temporal hierarchies and reconciliation

THieF can be understood in three steps: (*i*) construct non-overlapping temporally aggregate time series (levels), (*ii*) generate base forecasts at these levels independently, (*iii*) reconcile the base forecasts to combine the diverse information (Athanasopoulos et al., 2017). Temporal aggregation is a moving average that filters and strengthens different aspects of the original signal; therefore, the combination of the base forecasts helps recover information that would otherwise be difficult to estimate (Kourentzes et al., 2014).

The benefits of THieF primarily stem from three elements. First, it independently models different views of the time series, which is particularly advantageous when model misspecification is increased (Athanasopoulos et al., 2017). Second, it relies on an implicit forecast combination that leverages relationships among temporally aggregated series (Kourentzes & Athanasopoulos, 2019). Third, it uses restricted estimation of the implied combination weights, which reduces estimation uncertainty (Pritularga et al., 2021).

In Fig. 1, we plot the out-of-sample base and reconciled forecasts at four aggregation levels for a sample day, market, and model. THieF blends information from the base forecasts, and the relatively low bias of the average daily price forecast helps reduce bias at all other levels. The combination achieved by THieF is affine, but not convex (weights sum to unity but can be negative). This can help to improve forecasts further.

2.1. The summing matrix

Let m be the frequency of a time series, for instance, $m = 24$ for hourly data. Let $k = \{k_i\}_{i=1}^{p}$ be the divisors of m, ordered such that $m = k_p > k_{p-1} > \cdots > k_1 = 1$, and p is the number of factors. For example, for hourly data we have $k_i \in \{24, 12, 8, 6, 4, 3, 2, 1\}$ and $p = 8$. We define the *summing matrix* of size $(\sum_{i=1}^{p} k_i) \times m$ as:

$$\mathbf{S} = \left[\mathbf{S}_{k_p}^T \dots \mathbf{S}_{k_1}^T\right]^T, \tag{1}$$

where $\mathbf{S}_k^T$ is the transpose of $\mathbf{S}_k$. Each $\mathbf{S}_k$ is given by:

$$\mathbf{S}_k = s_k \mathbf{I}_{m/k} \otimes \mathbf{J}_{1,k}, \tag{2}$$

i.e., the weight $s_k = \frac{1}{k}$ multiplied by the Kronecker product between the identity matrix $\mathbf{I}_{m/k}$ of size $\frac{m}{k} \times \frac{m}{k}$

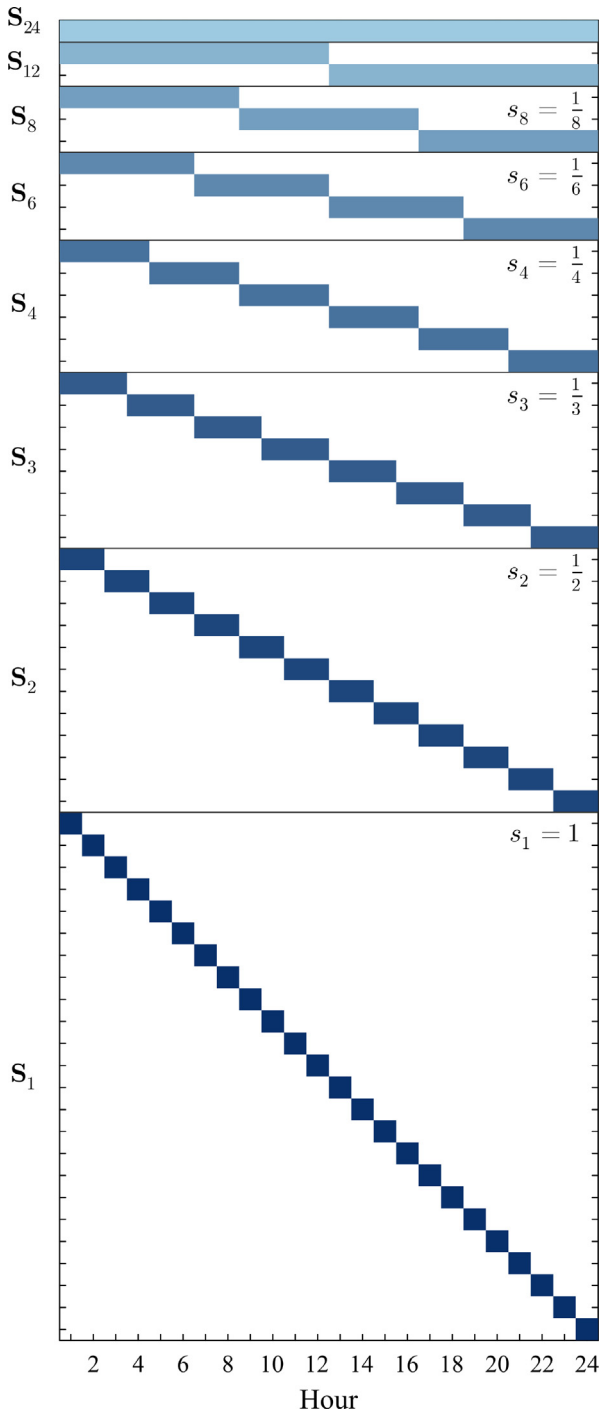


Fig. 2. The summing matrix $\mathbf{S}$ defined in Eq. (3). Blue blocks denote non-zero elements, and the shading indicates weights ranging from $s_{24} = \frac{1}{24}$ for $\mathbf{S}_{24}$ to $s_1 = 1$ for $\mathbf{S}_1$.

and a row vector $\mathbf{J}_{1,k}$ of size $1 \times k$ with all its elements equal to 1. For instance, for hourly data, we get:

$$\mathbf{S} = \left[\mathbf{S}_{24}^T\ \mathbf{S}_{12}^T\ \mathbf{S}_8^T\ \mathbf{S}_6^T\ \mathbf{S}_4^T\ \mathbf{S}_3^T\ \mathbf{S}_2^T\ \mathbf{S}_1^T\right]^T. \tag{3}$$

This is illustrated in Fig. 2. The top row $\mathbf{S}_{24}$ represents the baseload block, i.e., the average of the 24 hourly prices. The two rows of $\mathbf{S}_{12}$ correspond to the 12-hour blocks, respectively for hours 1-12 and hours 13-24. Analogously, the three rows of $\mathbf{S}_8$ map the hourly prices to the average price for the first, middle, and last 8-hourly blocks. The bottom-level $\mathbf{S}_1$ is a 24×24 identity matrix and corresponds to forecasts for the individual hours.

The temporal aggregation levels prescribed by k are non-overlapping for two reasons. First, overlapping temporal aggregation introduces serial autocorrelation that the base forecasting methods must capture. This not only increases model complexity but also obscures the identification of the underlying signal. Second, under the reasonable assumption that the base models may be misspecified, such serial autocorrelation can further complicate reconciliation in THieF.

2.2. Reconciliation

Let $\mathbf{p}_d$ be a column vector that contains the hourly observations for day d, then $\mathbf{P}_d = \mathbf{S}\mathbf{p}_d$ implements step (*i*). Further, let $\hat{\mathbf{P}}_\mathbf{d}$ be a vector of unreconciled base forecasts → step (*ii*). Following Athanasopoulos et al. (2017), THieF reconciliation is performed each day by computing:

$$\tilde{\mathbf{P}}_\mathbf{d} = \mathbf{S}(\mathbf{S}^T\mathbf{W}^{-1}\mathbf{S})^{-1}\mathbf{S}^T\mathbf{W}^{-1}\hat{\mathbf{P}}_\mathbf{d}, \tag{4}$$

where $\mathbf{W}$ is the covariance matrix of the base forecast errors, estimated on the errors from the training sample → step (*iii*). Since the covariance matrix for our hierarchical forecasts is 60×60, the total number of elements to estimate (1800) is larger than the sample size (1085; see Section 3). Therefore, we resort to regularized covariance matrix estimators.

Note that our formulation differs from Athanasopoulos et al. (2017) by using the mean for the aggregation, as in Kourentzes et al. (2014). This is due to the effect of weights s_k in the calculation of each $\mathbf{S}_k$ in Eq. (2). Effectively, this normalizes $\mathbf{W}$.

To see this, let $\mathbf{S}_\Sigma$ be the summing matrix as in Athanasopoulos et al. (2017), which is constructed using Eq. (1), but with all $s_k \equiv 1$. Further, we set $\mathbf{\Lambda} = diag(1/\mathbf{S}_\Sigma\mathbf{1})$ and use it to rescale sum-aggregates into mean-aggregates. Since $\mathbf{S} = \mathbf{\Lambda}\mathbf{S}_\Sigma$ and $\mathbf{\Lambda}^T = \mathbf{\Lambda}$, we can rewrite Eq. (4) as:

$$\tilde{\mathbf{P}}_\mathbf{d} = \mathbf{\Lambda}\mathbf{S}_\Sigma\left(\mathbf{S}_\Sigma^T\mathbf{\Lambda}\mathbf{W}^{-1}\mathbf{\Lambda}\mathbf{S}_\Sigma\right)^{-1}\mathbf{S}_\Sigma^T\mathbf{\Lambda}\mathbf{W}^{-1}\hat{\mathbf{P}}_\mathbf{d}. \tag{5}$$

Given that $\mathbf{W}_\Sigma^{-1} = (\mathbf{\Lambda}^{-1}\mathbf{W}\mathbf{\Lambda}^{-1})^{-1} = \mathbf{\Lambda}\mathbf{W}^{-1}\mathbf{\Lambda}$ and multiplying both sides by $\mathbf{\Lambda}^{-1}$ we obtain:

$$\mathbf{\Lambda}^{-1}\tilde{\mathbf{P}}_\mathbf{d} = \left[\mathbf{S}_\Sigma(\mathbf{S}_\Sigma^T\mathbf{W}_\Sigma^{-1}\mathbf{S}_\Sigma)^{-1}\mathbf{S}_\Sigma^T\mathbf{W}_\Sigma^{-1}\right]\mathbf{\Lambda}^{-1}\hat{\mathbf{P}}_\mathbf{d}. \tag{6}$$

The terms within the square brackets are identical to the reconciliation using $\mathbf{S}_\Sigma$ in Athanasopoulos et al. (2017). Therefore, the use of $\mathbf{S}$ instead of $\mathbf{S}_\Sigma$ normalizes $\mathbf{W}$, but otherwise results in the equivalent sum-aggregate forecasts.

2.3. Shrinkage

We considered three shrinkage approaches for estimating the covariance matrix $\mathbf{W}$, the constant correlation

model (Ledoit & Wolf, 2004), the diagonal shrinkage target (Schäfer & Strimmer, 2005), which has been shown to perform well in forecast reconciliation (Wickramasuriya et al., 2019), and the so-called 'variance scaling' (Athanasopoulos et al., 2017) that sets all off-diagonal elements to zero and corresponds to the Weighted Least Squares (WLS) estimator. We found that the first two approaches performed similarly well. The variance scaling also provided improvements, but was less impressive than the alternatives. In Section 5, we report the results for the less-conservative constant-correlation shrinkage of Ledoit and Wolf (2004). The results for the remaining approaches are provided in Appendix.

3. Datasets

To ensure a sound assessment of the THieF approach, we consider two major European power markets: EPEX-DE (Germany) and OMIE (Spain). Both datasets span seven years (05.01.2018-31.12.2024, see Fig. 3) and include publicly available data:

- three series specific to each market (hourly resolution) – day-ahead prices $p_{d,h}$, day-ahead load $\hat{L}_{d,h}$ and wind $\hat{W}_{d,h}$ generation forecasts ($\hat{W}_{d,h}$ is the sum of on- and off-shore generation; source: https://transparency.entsoe.eu),
- two series common to both markets (daily resolution) – the last known (hence the $d-2$ subscripts) closing prices of the nearest to delivery monthly coal API_{d-2} and yearly natural gas TTF_{d-2} futures (source: https://www.investing.com/).

Note that the EPEX-DE dataset is the same as used by Serafin and Weron (2025).

The first 1092 days (until 31.12.2020) are the initial training window. Note that, after accounting for the price lags (see Eq. (7) below), this corresponds to $1092 - 7 = 1085$ data points for training. Each day, the training window is rolled forward by 24 h.

The remaining 4-year period (starting 01.01.2021) is a challenging test set that includes the COVID-19 pandemic (lockdowns in Germany and Spain began in March 2020, with 15.12.2020 being the first day of hard lockdown in Germany), the Russian invasion of Ukraine (24.02.2022), and the soaring natural gas prices (Q4 2021-Q4 2022). Negative price spikes in Germany also mark this period. For instance, on Sunday, 02.07.2023, at hour 15, the price dropped to-$500 EUR/MWh due to low demand and high renewable generation. On the other hand, negative prices were not seen in the Spanish market before 2024, and even then, prices dropped only a few euros below zero. However, near-zero prices were reached about 5% of the time in 2024.

4. Computing base forecasts

4.1. Preliminaries

To demonstrate the versatility of our approach, we generate base forecasts using four models with four distinct architectures: linear regression, a shallow neural network, gradient boosting, and a transformer. The first two are often used as benchmarks in EPF (Billé et al., 2023; Ghelasi & Ziel, 2025; Hubicka et al., 2019; Janczura & Wójcik, 2022; Maciejowska et al., 2021; Serafin & Weron, 2025; Uniejewski & Maciejowska, 2023; Uniejewski et al., 2018; Ziel & Weron, 2018). The third has been reported to perform well in numerous forecasting competitions, including M5 (Januschowski et al., 2022). The fourth uses a transformer architecture, which is still rare in EPF (Bottieau et al., 2023). It is AutoGluon's new tabular foundation model, *Mitra*, which excels on datasets with fewer than 5000 samples and 100 features (Zhang et al., 2025).

All four models compute the electricity price forecast $\hat{p}_{d,h}$ for day d and block h as a function of 20 features, as in Serafin and Weron (2025):

$$\hat{p}_{d,h} = f\Big(p_{d-1,h}, \ldots, p_{d-7,h}, p^{min}_{d-1}, p^{max}_{d-1}, \hat{L}_{d,h}, \hat{W}_{d,h}, \text{API}_{d-2}, \text{TTF}_{d-2}, D^{(1)}_d, \ldots, D^{(7)}_d\Big), \tag{7}$$

where

- $p_{d-i,h}$ are the lagged prices of the same block in the last seven days $i = 1, \ldots, 7$,
- p^{min}_{d-1} and p^{max}_{d-1} are the minimum and maximum hourly prices of the previous day,
- $\hat{L}_{d,h}$ and $\hat{W}_{d,h}$ are the day-ahead load and wind generation forecasts for the target block,
- API_{d-2} and TTF_{d-2} are the closing prices of the nearest to delivery monthly coal (API2) and yearly natural gas futures (TTF) from day $d-2$, and
- $D^{(i)}_d$ are the weekday dummies.

All models are trained independently for each block h using a 3-year window of past values; each day, the window is rolled forward by one day. Overall, we consider 60 blocks: 24 × 1H (i.e., 24 one-hour blocks), 12 × 2H, 8 × 3H, 6 × 4H, 4 × 6H, 3 × 8H, 2 × 12H, and 1 × 24H (i.e., baseload).

For more robust parameter estimation, following Uniejewski et al. (2018) and Uniejewski (2026), before training the models, we preprocess the inputs using the *area hyperbolic sine* transformation: $\text{asinh}\big((y - \hat{\mu}_y)/\hat{\sigma}_y\big)$, where $\hat{\mu}_y$ and $\hat{\sigma}_y$ are the sample mean and sample standard deviation of y estimated on the training set. More precisely, $p_{d-i,h}$ for all $i = 1, \ldots, 7$ and each selected block h are transformed with a common $\hat{\mu}_y$ and $\hat{\sigma}_y$ calculated for the vector $[p_{d-1092,h}, \ldots, p_{d-1,h}]$. The extremes p^{min}_{d-1} and p^{max}_{d-1} as well as the exogenous variables $\hat{L}_{d,h}$, $\hat{W}_{d,h}$, API_{d-2} and TTF_{d-2} are transformed independently, also using 3-year vectors of past values; weekday dummies are not transformed.

4.2. AutoRegression with eXogenous inputs (ARX)

This expert – in the sense of Ziel and Weron (2018) – model is estimated via ordinary least squares (OLS) and uses the same inputs as Serafin and Weron (2025) to allow for direct comparisons. More precisely, the ARX model is given by:

$$\hat{p}_{d,h} = \sum_{i=1}^{7} \beta_i p_{d-i,h} + \beta_8 p^{min}_{d-1} + \beta_9 p^{max}_{d-1} + \beta_{10} \hat{L}_{d,h}$$

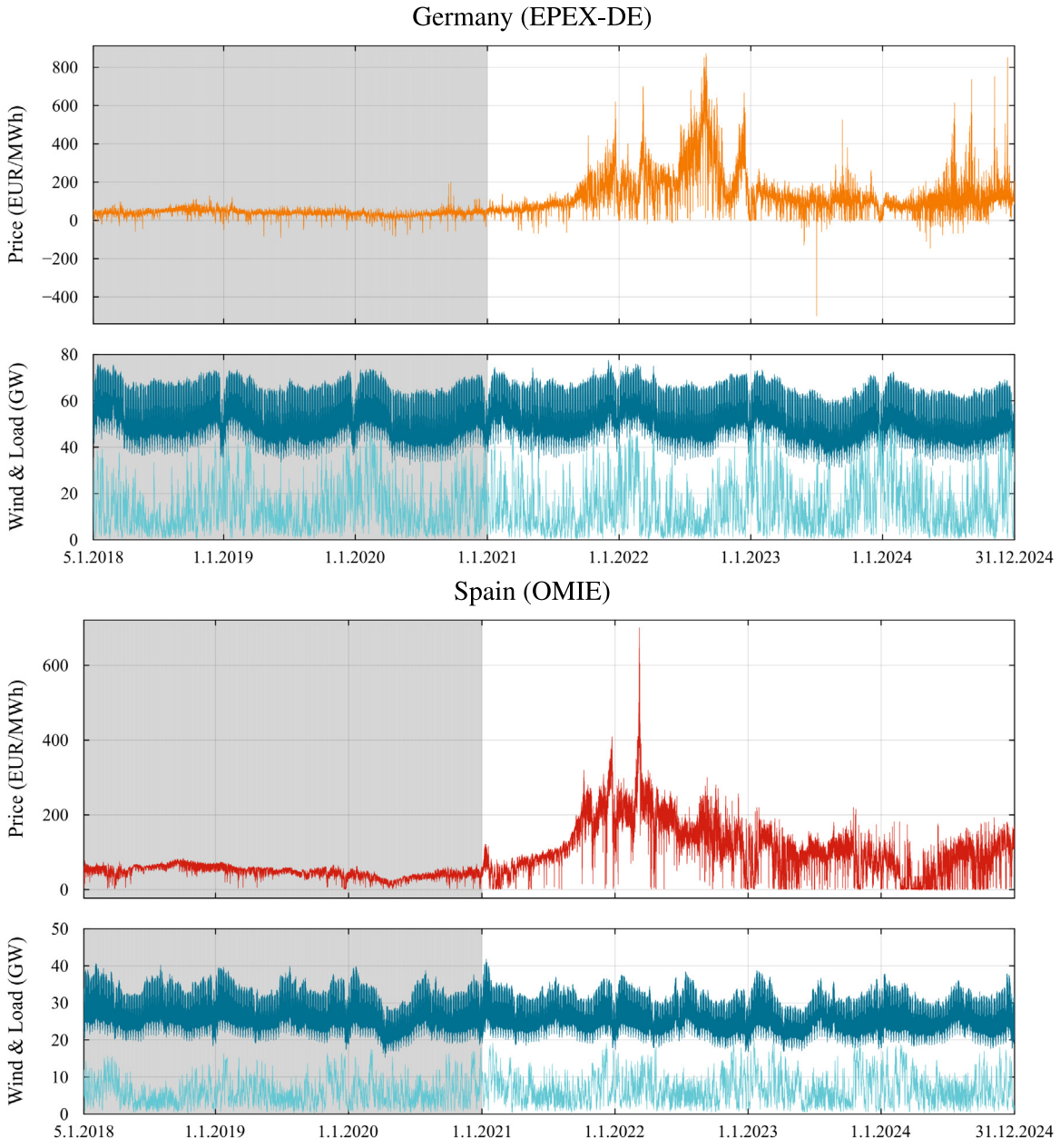


Fig. 3. German EPEX-DE (*top panels*) and Spanish OMIE (*bottom panels*) electricity market data: day-ahead prices (*orange* or *red*) and day-ahead load (*dark blue*) and wind generation (*light blue*) forecasts. The shaded area marks the first training window, i.e., the data used to train the models to generate forecasts for the first day of the test period (1.1.2021).

$$+ \beta_{11}\hat{W}_{d,h} + \beta_{12}\mathrm{API}_{d-2} + \beta_{13}\mathrm{TTF}_{d-2} + \sum_{i=14}^{20} \beta_i D_d^{(i)}. \tag{8}$$

4.3. Nonlinear ARX (NARX)

The nonlinear counterpart of ARX approximates $f(\cdot)$ in Eq. (7) with a shallow feedforward neural network using the series-parallel architecture (Xie et al., 2009). As in Hubicka et al. (2019), the hidden layer consists of 5 neurons and uses hyperbolic tangent activation, with a linear function in the output layer. The weights are calculated in Matlab R2025a using the Levenberg–Marquardt algorithm with early stopping based on a 10% validation set. To mitigate the effects of parameter estimation uncertainty, we clip the network output to $[-3, 3]$ before applying the inverse transformation (i.e., the hyperbolic sine, see Section 4.1) and use a committee machine of 10 networks (Marcjasz et al., 2019). More precisely, the final $\hat{p}_{d,h}$ is obtained by training the network 10 times for each d and h, and averaging the 10 price forecasts; no hyperparameter optimization is performed.

4.4. EXtreme Gradient Boosting (XGB)

The third model is an ensemble of gradient-boosted decision trees (GBDT; Hastie et al., 2009). We use eXtreme GBDT implemented in the Python package XGBoost v3.0 (Chen & Guestrin, 2016), with the mean squared error (MSE) as the loss function for training and hyperparameter tuning. The latter is carried out independently for each of the 60 blocks (see Section 4.1) using Bayesian sequential optimization (Akiba et al., 2019) on the initial 3-year training window (05.01.2018-31.12.2024; see Fig. 3) with a random 10% left out for validation. We select from the following set of hyperparameters: maximum depth of a single tree (between 2 and 10), learning rate (between 10^{-4} and 1 on log scale), subsample ratio of the training instances (between $\frac{1}{2}$ and 1), minimum sum of instance weight required for partitioning a leaf node (between 0 and 10), minimum loss reduction required for partioning a leaf node (between 0 and $\frac{1}{2}$), strengths of the L1 and L2 regularization on model weights (each between 10^{-3} and 10 on log scale). Additionally, we apply early stopping to limit the number of boosting rounds (capped at 1000). Since the hyperparameter selection process is stochastic, we perform it 10 times, and the final $\hat{p}_{d,h}$ is obtained by averaging the forecasts generated from 10 XGB instances with different hyperparameter sets.

4.5. Mitra

The final forecaster is Amazon's state-of-the-art tabular foundation model, *Mitra*, released with AutoGluon v1.4 in July 2025, based on a 12-layer, 72-million-parameter transformer architecture (Zhang et al., 2025). Since it is pre-trained on purely synthetic data, evaluating its performance on historic time series does not pose data contamination issues. Although Mitra is not specifically tailored to time series, our forecasting task requires only one-step-ahead predictions, with Eq. (7) being equivalent to a tabular regression problem.

We generate forecasts in *zero-shot* mode, i.e., without optimizing hyperparameters and relying on a pre-trained model specialized for regression tasks. We fit Mitra using AutoGluon's *TabularPredictor* class with fine-tuning turned off and other settings kept at their defaults. Mitra is based on the in-context learning paradigm; hence, the training set serves as support examples that condition the prediction. A single forecast $\hat{p}_{d,h}$ is generated by passing input data, see Eq. (7), as a query. As with the other models considered in our study, we use Mitra in a rolling window approach and train one model for each day d and block h in the test period.

5. Empirical results

To measure differences in predictive performance, we compute the mean absolute errors (MAE) and root mean squared errors (RMSE) for the four models presented in Section 4 across both markets and over the entire 4-year test period. The errors and the respective percentage gains from forecast reconciliation are reported in Table 1. Additionally, the gains are plotted in Fig. 4. Clearly, THieF improves forecast accuracy across all aggregation levels, for every considered base model and for both markets.

The relative gains in the German (EPEX-DE) market are generally greater than in the Spanish (OMIE) market. The highest MAE accuracy improvements are obtained with the XGB model, with a 6.7% average gain in Germany and a 4.8% in Spain. XGB also achieves the highest average reconciliation gains in RMSE, with improvements of 7.6% and 5.0%, respectively. Additionally, gains are typically higher for larger blocks, which is particularly relevant given the quadrupling of the number of products in the European day-ahead market and the resulting importance of block trading (European Commission, 2025).

Importantly, all reconciled forecasts are significantly more accurate than the corresponding unreconciled ones, as measured by the multivariate variant of the Diebold–Mariano test (Lago et al., 2021; Ziel & Weron, 2018) based on the L2-norm of forecast errors. For the EPEX-DE market, the test indicated significance at the 1% level for each aggregation level (1H, 2H, 3H, 4H, 6H, 8H, 12H, 24H) and model. The same holds for the OMIE market, except for three isolated cases: 1H NARX, 1H Mitra, and 24H NARX, for which the differences in unconditional predictive ability were significant at the 5% level.

The fact that the results are consistent across architectures ranging from a parsimonious linear regression with 20 parameters to a state-of-the-art transformer with up to 72 million parameters is a strong argument for using the THieF approach in day-ahead EPF. Moreover, the computational overhead of implementing THieF in a forecasting pipeline scales linearly with the baseline. The cost of the reconciliation itself is negligible; hence, the only relevant cost is the cost of estimating models at additional aggregation levels. Overall, THieF increases the computational cost to about 2.5 times that of the 24-hour baseline. This increase can be considered a minor overhead in the context of day-ahead electricity price forecasting, since the per-day runtime for a single ARX, NARX, or XGB model is sub-second on a consumer-grade CPU. In contrast, Mitra requires roughly one second on an NVIDIA RTX A600 GPU.

Additionally, to analyze the reconciliation gains not only across different models but also across market regimes, Fig. 4 provides a yearly breakdown of the RMSE-based gains reported in Table 1. Notably, THieF delivers accuracy gains in the German EPEX market across all years and blocks for ARX, XGB, and Mitra. For NARX, there are a few instances where the reconciled forecasts are less accurate than the unreconciled ones, particularly for hourly forecasts in 2023–2024 and for 3-, 4-, and 8-hour blocks in 2024.

The results for the Spanish OMIE are less robust. While XGB is consistently improved by THieF in each year considered, ARX does not benefit in 2022 and loses accuracy in 2023. Similarly, the advantage of THieF is less evident for NARX in 2022 and for Mitra in 2023–2024. These findings may reflect the relevance of historical price dynamics from the 2022 energy crisis for forecasting future prices in these two markets. Note that Spanish electricity prices exhibit lower volatility from 2022Q3 onward: after the significant spikes in the first half of 2022, prices remained

Table 1
Mean absolute errors and root mean squared errors of the base (MAE, RMSE) and reconciled (MAE_R, $RMSE_R$) forecasts with the respective gains from reconciliation (%) for all considered hierarchy levels over the 4-year test period in the German (*left*) and Spanish (*right*) markets. All reconciled forecasts are significantly more accurate than the corresponding unreconciled ones, as measured by the multivariate variant of the Diebold–Mariano test (Lago et al., 2021; Ziel & Weron, 2018) based on the L2-norm of forecast errors; see text for details.

		Germany (EPEX-DE)						Spain (OMIE)					
Level	**Model**	**MAE**	**MAE_R**	**%**	**RMSE**	**$RMSE_R$**	**%**	**MAE**	**MAE_R**	**%**	**RMSE**	**$RMSE_R$**	**%**
1H	ARX	26.93	26.17	2.8	39.83	38.75	2.7	16.72	16.57	0.9	24.40	24.38	0.1
	NARX	23.06	22.29	3.3	35.62	34.63	2.8	16.57	16.24	2.0	24.07	23.92	0.6
	XGB	23.33	22.30	4.4	37.05	35.02	5.5	16.54	15.98	3.4	24.57	23.67	3.7
	Mitra	21.37	21.11	1.2	33.31	32.91	1.2	15.92	15.84	0.6	23.61	23.20	1.7
2H	ARX	26.44	25.68	2.9	38.95	37.86	2.8	16.41	16.25	1.0	23.96	23.92	0.2
	NARX	22.64	21.61	4.5	34.84	33.44	4.0	16.29	15.80	3.0	23.71	23.29	1.8
	XGB	22.74	21.66	4.8	35.87	33.88	5.6	16.19	15.56	3.9	24.11	23.09	4.2
	Mitra	20.92	20.38	2.6	32.43	31.60	2.6	15.56	15.38	1.1	23.17	22.54	2.7
3H	ARX	26.11	25.35	2.9	38.28	37.17	2.9	16.16	16.01	1.0	23.60	23.52	0.3
	NARX	22.12	21.17	4.3	33.89	32.63	3.7	16.01	15.54	2.9	23.44	22.86	2.5
	XGB	22.26	21.24	4.6	35.10	33.09	5.7	15.93	15.27	4.2	23.75	22.67	4.5
	Mitra	20.61	19.90	3.5	31.87	30.69	3.7	15.31	15.10	1.3	22.72	22.09	2.8
4H	ARX	25.65	24.85	3.1	37.50	36.33	3.1	15.88	15.71	1.1	23.25	23.17	0.4
	NARX	21.75	20.66	5.0	33.27	31.75	4.6	15.68	15.27	2.6	22.96	22.48	2.1
	XGB	22.01	20.72	5.8	34.24	32.19	6.0	15.78	14.95	5.3	23.60	22.26	5.7
	Mitra	20.13	19.39	3.7	30.86	29.79	3.5	15.01	14.79	1.4	22.19	21.70	2.2
6H	ARX	25.14	24.35	3.1	36.42	35.30	3.1	15.60	15.45	1.0	22.77	22.64	0.6
	NARX	21.08	20.00	5.1	32.00	30.52	4.6	15.27	14.86	2.7	22.41	21.93	2.2
	XGB	21.52	20.15	6.4	33.38	31.05	7.0	15.41	14.60	5.3	22.93	21.74	5.2
	Mitra	19.51	18.77	3.8	29.74	28.48	4.2	14.77	14.39	2.6	21.87	21.12	3.4
8H	ARX	24.37	23.65	2.9	35.59	34.35	3.5	15.14	14.99	1.0	22.12	21.99	0.6
	NARX	20.54	19.42	5.4	31.20	29.71	4.8	15.01	14.52	3.2	21.92	21.35	2.6
	XGB	21.06	19.63	6.8	32.94	30.24	8.2	15.10	14.18	6.1	22.50	21.08	6.3
	Mitra	19.45	18.22	6.3	29.34	27.74	5.5	14.44	14.01	3.0	21.45	20.47	4.5
12H	ARX	22.81	22.14	2.9	32.68	31.79	2.7	14.27	14.22	0.4	20.83	20.77	0.3
	NARX	19.37	17.80	8.1	28.96	27.07	6.5	14.25	13.80	3.1	20.64	20.23	2.0
	XGB	19.70	17.95	8.9	30.55	27.55	9.8	14.29	13.49	5.6	21.24	19.99	5.9
	Mitra	17.59	16.57	5.8	26.27	24.81	5.6	13.67	13.28	2.8	20.16	19.39	3.9
24H	ARX	20.92	20.50	2.0	30.28	29.27	3.4	13.57	13.40	1.2	19.50	19.39	0.6
	NARX	17.99	16.53	8.2	27.24	25.08	7.9	13.29	12.94	2.6	19.14	18.98	0.9
	XGB	18.80	16.59	11.7	29.50	25.54	13.4	13.25	12.58	5.1	19.61	18.66	4.8
	Mitra	16.56	15.22	8.1	24.81	22.70	8.5	12.91	12.42	3.8	18.84	18.11	3.8

below 300 EUR/MWh, with only a few spikes above 200 EUR/MWh in 2023–2024. After the extreme volatility during the crisis, the Spanish market entered a relatively calmer regime. In contrast, large price spikes were still observed in the EPEX market even in the final days of 2024.

6. Conclusions

In this study, we demonstrated the benefits of *temporal hierarchy forecasting* (THieF) for predicting electricity prices. We demonstrated this for the German and Spanish power markets, using a variety of base-forecasting methods with increasing complexity and diverse modeling assumptions. We found a consistent and statistically significant improvement from reconciliation in all setups – up to 5.5% for hourly and 13.4% for baseload prices – with gains increasing at higher aggregation levels (→ larger blocks).

The latter is particularly relevant given the quadrupling of the number of products (European Commission, 2025). On 30 September 2025, the European day-ahead electricity markets moved from hourly (→ 24 products per day) to 15-minute trading intervals (→ 96 products per day). Although this finer granularity offers a more accurate representation of expected generation and demand, it also adds complexity to electricity trading, thereby increasing the importance of block trading and temporal reconciliation.

In principle, if the base forecasts were not misspecified, there would be limited scope for THieF to improve accuracy. However, in practice, any forecasting method will have some degree of misspecification relative to the unknown underlying electricity price-generating process. THieF augments these base forecasts, enabling them to better capture the short- and long-term dynamics present in the time series. An interesting finding is that the improvements are consistent for all base forecasts, including Mitra. The foundation model's misspecification is arguably of a different nature because it is used in a zero-shot, pre-trained mode that does not involve estimating model parameters. Moreover, our evaluation provides additional evidence of the usefulness of THieF when base forecasts use covariates, with most applications in the literature focusing on the univariate case.

There are various extensions of this work. We applied the same base forecast to all aggregation levels; however, a more diverse model selection could be beneficial, better leveraging the properties of each level and thus minimizing computational overhead. Likewise, considering a richer set of base models would be interesting, including the well-performing in EPF applications LASSO-Estimated AutoRegressions (LEAR; Lago et al., 2021) and Distributional Deep Neural Networks (DDNN; Marcjasz



Fig. 4. Percentage gains in RMSE from forecast reconciliation in the German EPEX (*dark blue*) and Spanish OMIE (*light blue*) markets for block sizes ranging from 1 to 24 h, separately for each of the four models (*left to right*) and calendar years (*top to bottom*).

et al., 2023). Furthermore, investigating the impact of temporal reconciliation on probabilistic EPF is a useful extension. Beyond any accuracy gains, THieF provides reconciled forecasts across all levels, which can support the alignment of different decisions and potentially new strategies in energy markets (Maciejowska & Nitka, 2026). Finally, the temporal hierarchy was restricted to using aggregation levels based solely on factors of m. Unequal aggregation blocks are relevant to decisions supported by EPF, and embedding them in THieF can be beneficial for both forecasting performance and decision-making. These are fruitful avenues for future research.

CRediT authorship contribution statement

Arkadiusz Lipiecki: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Kaja Bilińska:** Writing – original draft, Software, Investigation. **Nikolaos Kourentzes:** Writing – review & editing, Writing – original draft, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Rafał Weron:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The study was partially supported by the National Science Center (NCN, Poland) through grant no. 2018/30/A/HS4/00444 (to AL and KB), by the National Science Center (NCN, Poland) and the German Research Foundation (DFG, Germany) through grant no. 2021/43/I/HS4/02578 (to RW; DFG no. 505565850), and by Riksbankens Jubileumsfond (Sweden) through grant no. SAB22-0073 (to NK).

Appendix. Results for alternative covariance matrix approximations

In Table A.2 we provide the results for the diagonal target shrinkage of Schäfer and Strimmer (2005). The results are very similar to those in Table 1 for the constant correlation shrinkage of Ledoit and Wolf (2004). For the EPEX-DE market, the multivariate variant of the Diebold–Mariano test (Lago et al., 2021; Ziel & Weron, 2018) indicated significance at the 1% level for each aggregation level (1H, 2H, 3H, 4H, 6H, 8H, 12H, 24H) and model. The same holds for the OMIE market, except for three isolated cases. However, this time the improvements in predictive ability for 1H NARX, 1H Mitra, and 24H NARX are not significant at the 5% level.

In Table A.3 we provide the results for the so-called 'variance scaling' (Athanasopoulos et al., 2017) that sets all off-diagonal elements to zero and corresponds to the WLS estimator. Clearly, in this case, the gains from reconciliation are smaller, and in a few instances, accuracy losses can be observed. This indicates that accounting for error correlations is necessary for effective reconciliation in day-ahead EPF.

Data and code availability

The datasets and sources are described in Section 3. The replication package is publicly available at https://github.com/lipiecki/thief.

Table A.2
Results for forecast reconciliation based on the diagonal target shrinkage of Schäfer and Strimmer (2005). Like in Table 1, we report mean absolute errors and root mean squared errors of the base (MAE, RMSE) and reconciled (MAE_R, $RMSE_R$) forecasts with the respective gains from reconciliation (%) for all considered hierarchy levels over the 4-year test period in the German (*left*) and Spanish (*right*) markets.

		Germany (EPEX-DE)						Spain (OMIE)					
Level	**Model**	**MAE**	**MAE_R**	**%**	**RMSE**	**$RMSE_R$**	**%**	**MAE**	**MAE_R**	**%**	**RMSE**	**$RMSE_R$**	**%**
1H	ARX	26.93	26.08	3.1	39.83	38.56	3.2	16.72	16.52	1.2	24.4	24.28	0.5
	NARX	23.06	22.27	3.4	35.62	34.56	3.0	16.57	16.34	1.4	24.07	24.02	0.2
	XGB	23.33	22.36	4.2	37.05	35.13	5.2	16.54	16	3.2	24.57	23.7	3.5
	MITRA	21.37	21.08	1.4	33.31	32.84	1.4	15.92	15.88	0.3	23.61	23.25	1.5
2H	ARX	26.44	25.6	3.2	38.95	37.66	3.3	16.41	16.2	1.2	23.96	23.82	0.6
	NARX	22.64	21.59	4.6	34.84	33.37	4.2	16.29	15.89	2.4	23.71	23.39	1.4
	XGB	22.74	21.72	4.5	35.87	33.99	5.3	16.19	15.59	3.7	24.11	23.13	4.1
	MITRA	20.92	20.34	2.8	32.43	31.52	2.8	15.56	15.43	0.8	23.17	22.59	2.5
3H	ARX	26.11	25.26	3.2	38.28	36.98	3.4	16.16	15.96	1.3	23.59	23.43	0.7
	NARX	22.12	21.14	4.4	33.89	32.55	4.0	16.01	15.63	2.3	23.44	22.96	2.1
	XGB	22.26	21.3	4.3	35.1	33.2	5.4	15.93	15.29	4.0	23.75	22.71	4.4
	MITRA	20.61	19.86	3.7	31.87	30.61	3.9	15.31	15.14	1.1	22.72	22.14	2.6
4H	ARX	25.65	24.76	3.4	37.5	36.13	3.6	15.88	15.67	1.3	23.25	23.07	0.8
	NARX	21.75	20.6	5.3	33.27	31.67	4.8	15.68	15.36	2.1	22.96	22.57	1.7
	XGB	22.01	20.78	5.6	34.24	32.29	5.7	15.78	14.97	5.2	23.6	22.29	5.5
	MITRA	20.13	19.34	3.9	30.86	29.7	3.8	15.01	14.84	1.1	22.19	21.75	2.0
6H	ARX	25.14	24.28	3.4	36.42	35.12	3.6	15.6	15.39	1.3	22.77	22.55	1.0
	NARX	21.08	19.94	5.4	32	30.42	4.9	15.27	14.95	2.1	22.41	22.02	1.8
	XGB	21.52	20.2	6.2	33.38	31.15	6.7	15.41	14.61	5.2	22.93	21.77	5.0
	MITRA	19.51	18.71	4.1	29.73	28.38	4.6	14.77	14.44	2.2	21.87	21.18	3.1
8H	ARX	24.37	23.59	3.2	35.59	34.15	4.0	15.14	14.95	1.3	22.12	21.9	1.0
	NARX	20.54	19.39	5.6	31.2	29.61	5.1	15	14.61	2.6	21.92	21.42	2.3
	XGB	21.06	19.68	6.6	32.94	30.34	7.9	15.1	14.2	6.0	22.5	21.12	6.2
	MITRA	19.45	18.18	6.5	29.34	27.63	5.8	14.44	14.06	2.6	21.45	20.53	4.3
12H	ARX	22.81	22.1	3.1	32.68	31.64	3.2	14.27	14.19	0.6	20.83	20.68	0.7
	NARX	19.37	17.72	8.5	28.95	26.94	7.0	14.25	13.87	2.7	20.64	20.28	1.7
	XGB	19.7	17.98	8.7	30.55	27.61	9.6	14.29	13.5	5.5	21.24	20.01	5.8
	MITRA	17.59	16.5	6.2	26.27	24.69	6.0	13.67	13.34	2.4	20.16	19.44	3.6
24H	ARX	20.92	20.52	1.9	30.28	29.15	3.8	13.57	13.36	1.5	19.5	19.31	1.0
	NARX	17.99	16.41	8.8	27.24	24.9	8.6	13.29	13	2.1	19.14	19.01	0.7
	XGB	18.8	16.63	11.5	29.5	25.6	13.2	13.25	12.57	5.1	19.61	18.66	4.8
	MITRA	16.56	15.15	8.5	24.81	22.56	9.1	12.91	12.43	3.7	18.84	18.17	3.6

Table A.3
Results for the so-called 'variance scaling' (Athanasopoulos et al., 2017) that sets all off-diagonal elements to zero and corresponds to the WLS estimator. Like in Tables 1 and A.2, we report mean absolute errors and root mean squared errors of the base (MAE, RMSE) and reconciled (MAE_R, $RMSE_R$) forecasts with the respective gains from reconciliation (%) for all considered hierarchy levels over the 4-year test period in the German (*left*) and Spanish (*right*) markets.

		Germany (EPEX-DE)						Spain (OMIE)					
Level	**Model**	**MAE**	**MAE_R**	**%**	**RMSE**	**$RMSE_R$**	**%**	**MAE**	**MAE_R**	**%**	**RMSE**	**$RMSE_R$**	**%**
1H	ARX	26.93	26.92	0.0	39.83	39.82	0.0	16.72	16.72	0.0	24.4	24.43	-0.1
	NARX	23.06	22.99	0.3	35.62	35.55	0.2	16.57	16.49	0.5	24.07	24.03	0.2
	XGB	23.33	23.25	0.3	37.05	36.82	0.6	16.54	16.51	0.2	24.57	24.53	0.2
	MITRA	21.37	21.42	-0.2	33.31	33.38	-0.2	15.92	15.88	0.3	23.61	23.54	0.3
2H	ARX	26.44	26.42	0.1	38.95	38.93	0.0	16.41	16.4	0.0	23.96	23.96	0.0
	NARX	22.64	22.29	1.5	34.84	34.36	1.4	16.29	16.06	1.4	23.71	23.39	1.4
	XGB	22.74	22.63	0.5	35.87	35.73	0.4	16.19	16.1	0.6	24.11	23.96	0.6
	MITRA	20.92	20.64	1.3	32.43	32.01	1.3	15.56	15.39	1.1	23.17	22.84	1.4
3H	ARX	26.11	26.07	0.1	38.28	38.24	0.1	16.16	16.15	0.1	23.59	23.57	0.1
	NARX	22.12	21.88	1.1	33.89	33.55	1.0	16.01	15.79	1.4	23.44	22.96	2.0
	XGB	22.26	22.21	0.2	35.1	34.97	0.4	15.93	15.8	0.8	23.75	23.56	0.8
	MITRA	20.61	20.19	2.1	31.87	31.09	2.4	15.31	15.09	1.4	22.72	22.38	1.5
4H	ARX	25.65	25.59	0.2	37.5	37.41	0.2	15.88	15.86	0.1	23.25	23.21	0.2
	NARX	21.75	21.37	1.7	33.27	32.68	1.8	15.68	15.49	1.2	22.96	22.58	1.6
	XGB	22.01	21.74	1.2	34.24	34.13	0.3	15.78	15.5	1.8	23.6	23.17	1.8
	MITRA	20.13	19.64	2.4	30.86	30.16	2.3	15.01	14.8	1.4	22.19	21.99	0.9
6H	ARX	25.14	25.05	0.4	36.42	36.3	0.3	15.6	15.54	0.4	22.77	22.67	0.5
	NARX	21.08	20.79	1.4	32	31.45	1.7	15.27	15.13	1.0	22.41	22.03	1.7
	XGB	21.52	21.18	1.6	33.38	33	1.1	15.41	15.16	1.6	22.93	22.66	1.2
	MITRA	19.51	18.99	2.7	29.73	28.84	3.0	14.77	14.39	2.5	21.87	21.39	2.2
8H	ARX	24.37	24.33	0.2	35.59	35.44	0.4	15.14	15.07	0.4	22.12	22.01	0.5
	NARX	20.54	20.16	1.8	31.2	30.66	1.8	15	14.77	1.5	21.92	21.45	2.1
	XGB	21.06	20.62	2.1	32.94	32.25	2.1	15.1	14.74	2.4	22.5	22.04	2.0
	MITRA	19.45	18.47	5.0	29.34	28.07	4.3	14.44	13.97	3.2	21.45	20.74	3.3
12H	ARX	22.81	22.79	0.1	32.68	32.72	-0.1	14.27	14.26	0.1	20.83	20.75	0.4
	NARX	19.37	18.66	3.7	28.95	28.03	3.2	14.25	14.06	1.3	20.64	20.35	1.4
	XGB	19.7	19.08	3.2	30.55	29.8	2.4	14.29	14.05	1.7	21.24	20.99	1.2
	MITRA	17.59	16.74	4.8	26.27	25.04	4.7	13.67	13.24	3.2	20.16	19.63	2.6
24H	ARX	20.92	21	-0.4	30.28	30.12	0.6	13.57	13.42	1.1	19.5	19.34	0.8
	NARX	17.99	17.37	3.5	27.24	26.13	4.1	13.29	13.19	0.7	19.14	19.1	0.2
	XGB	18.8	17.77	5.4	29.5	27.93	5.3	13.25	13.21	0.3	19.61	19.76	-0.8
	MITRA	16.56	15.36	7.3	24.81	22.92	7.6	12.91	12.36	4.2	18.84	18.34	2.7

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